

3 Data Analysis

3.1 Results from Eight Cities

This section presents the results of the multivariate nonlinear polynomial modeling applied to each of the eight selected cities. The analysis examines the strength and nature of the relationship between NDVI and LST within the predefined rectangular study areas. For each city, a range of polynomial models was systematically fitted, and the optimal model was identified based on the adjusted R^2 , which reflects the best overall explanatory power while accounting for model complexity. The resulting equations are compared across cities to assess spatial similarities and differences in the NDVI-LST relationship. Furthermore, t-tests were conducted on individual coefficients to evaluate their statistical significance, and a cross-city regression comparison was performed to identify common patterns as well as city-specific variations in the vegetation-temperature interaction.

3.1.1 Shanghai

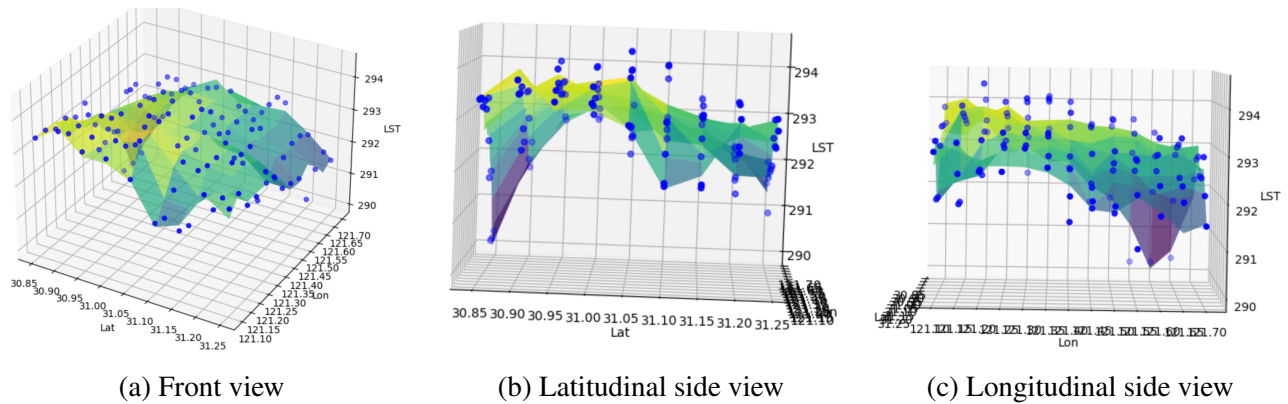


Figure 3: Three views of NDVI-LST Model of Shanghai

For Shanghai, the polynomial regression model that achieved the highest adjusted coefficient of determination was a third-degree model, with an adjusted R^2_{adj} of 0.6219. This indicates that approximately 62.2% of the variance in LST can be explained by the selected combination of NDVI,

longitude, latitude, and their higher-order interaction terms. The final fitted equation is as follows:

$$\begin{aligned} \text{LST} = & 391.728890 + 128647.837104 \text{NDVI} + 100953.586406 \text{Lon} - 396393.868389 \text{Lat} \\ & - 27504.541474 \text{NDVI}^2 + 4973.030846 \text{Lon} \cdot \text{NDVI} - 1956.270857 \text{Lon}^2 \\ & - 27250.384131 \text{Lat} \cdot \text{NDVI} + 8748.785027 \text{Lat} \cdot \text{Lon} - 4194.741648 \text{Lat}^2 \\ & + 412.969009 \text{NDVI}^3 + 123.384292 \text{Lon} \cdot \text{NDVI}^2 - 39.808041 \text{Lon}^2 \cdot \text{NDVI} \\ & + 6.652120 \text{Lon}^3 + 390.462883 \text{Lat} \cdot \text{NDVI}^2 + 149.210242 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\ & - 14.675540 \text{Lat} \cdot \text{Lon}^2 + 143.267676 \text{Lat}^2 \cdot \text{NDVI} - 84.113713 \text{Lat}^2 \cdot \text{Lon} \\ & + 153.936435 \text{Lat}^3 \end{aligned}$$

3.1.2 Suzhou

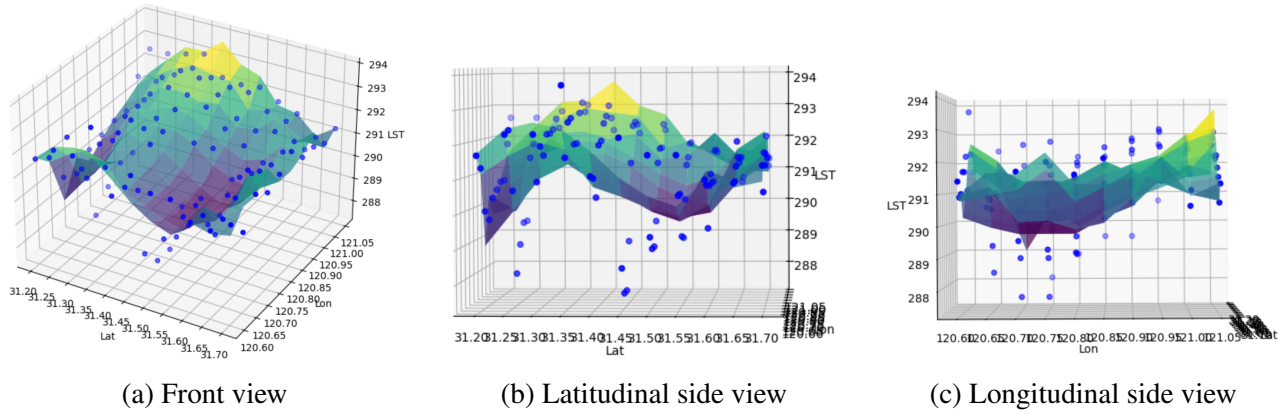


Figure 4: Three views of NDVI-LST Model of Suzhou

For Suzhou, the third-degree polynomial model provided the best fit, with an adjusted coefficient of determination of 0.3688. This suggests that approximately 36.9% of the variation in LST was explained by NDVI, geographic coordinates, and their nonlinear and interaction terms. The final fitted equation is as follows:

$$\begin{aligned} \text{LST} = & 61709.109441 - 5146316.074043 \text{NDVI} - 49646.817509 \text{Lon} + 253489.072961 \text{Lat} \\ & - 12556.405427 \text{NDVI}^2 + 77102.312153 \text{Lon} \cdot \text{NDVI} + 222.865196 \text{Lon}^2 \\ & + 31367.636206 \text{Lat} \cdot \text{NDVI} + 401.177304 \text{Lat} \cdot \text{Lon} - 9010.007744 \text{Lat}^2 \\ & + 90.185615 \text{NDVI}^3 + 38.348388 \text{Lon} \cdot \text{NDVI}^2 - 297.095366 \text{Lon}^2 \cdot \text{NDVI} \\ & - 0.996674 \text{Lon}^3 + 249.101070 \text{Lat} \cdot \text{NDVI}^2 - 169.808002 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\ & + 8.474057 \text{Lat} \cdot \text{Lon}^2 - 175.498654 \text{Lat}^2 \cdot \text{NDVI} - 37.979818 \text{Lat}^2 \cdot \text{Lon} \\ & + 144.821238 \text{Lat}^3 \end{aligned}$$

3.1.3 Jiaxing

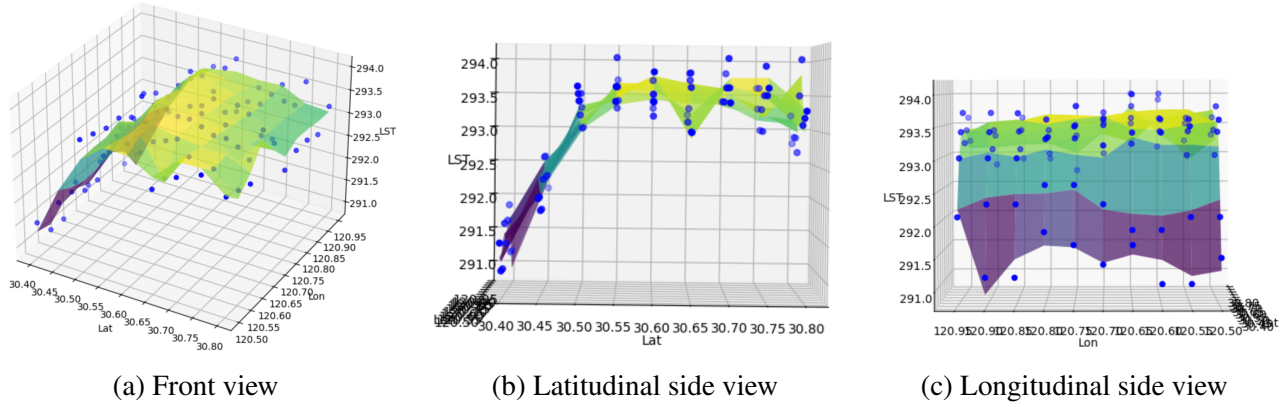


Figure 5: Three views of NDVI-LST Model of Jiaxing

For Jiaxing, the best-fitting model was a third-degree polynomial, achieving an adjusted coefficient of determination of 0.8468. This indicates that approximately 84.7% of the variance in LST was explained by the combination of NDVI, longitude, latitude, and their higher-order interaction terms, suggesting a strong and well-captured relationship between surface temperature and urban vegetation in the region. The final fitted equation is presented below:

$$\begin{aligned}
 \text{LST} = & -4099.388188 + 320465.789908 \text{NDVI} - 142155.232845 \text{Lon} + 554354.808392 \text{Lat} \\
 & - 17379.038276 \text{NDVI}^2 - 6328.058278 \text{Lon} \cdot \text{NDVI} + 1763.992584 \text{Lon}^2 \\
 & + 4508.401459 \text{Lat} \cdot \text{NDVI} - 4514.798645 \text{Lat} \cdot \text{Lon} - 9210.079458 \text{Lat}^2 \\
 & + 33.714438 \text{NDVI}^3 + 160.449121 \text{Lon} \cdot \text{NDVI}^2 + 30.547153 \text{Lon}^2 \cdot \text{NDVI} \\
 & - 6.717919 \text{Lon}^3 - 67.257831 \text{Lat} \cdot \text{NDVI}^2 - 38.703394 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\
 & + 21.339348 \text{Lat} \cdot \text{Lon}^2 + 3.563681 \text{Lat}^2 \cdot \text{NDVI} - 10.157343 \text{Lat}^2 \cdot \text{Lon} \\
 & + 113.338094 \text{Lat}^3
 \end{aligned}$$

Compared with other cities, the model for Jiaxing demonstrates the strongest predictive power. The surface exhibits a relatively smooth shape, with a clear decreasing trend from north to south, suggesting a clearer and more robust spatial relationship between vegetation and LST in this city.

3.1.4 Wuhu

For Wuhu, the third-degree polynomial model was identified as the best fit, achieving an adjusted coefficient of determination of 0.5048. This indicates that approximately 50.5% of the variance in LST was explained by the model, reflecting a moderate relationship between surface temperature and

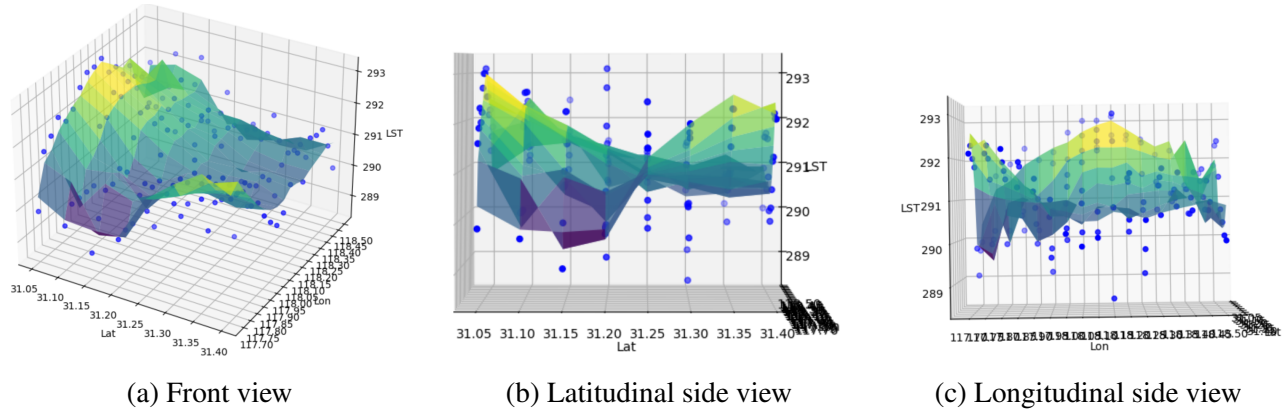


Figure 6: Three views of NDVI-LST Model of Wuhu

the predictor variables (NDVI, longitude, latitude, and their interactions). The model captures the nonlinear nature of these influences while avoiding overfitting.

$$\begin{aligned}
 \text{LST} = & 3265.119717 - 407641.756443 \text{NDVI} - 123416.556398 \text{Lon} + 470514.253535 \text{Lat} \\
 & - 6563.635176 \text{NDVI}^2 + 8772.909316 \text{Lon} \cdot \text{NDVI} + 2466.449869 \text{Lon}^2 \\
 & - 6865.084521 \text{Lat} \cdot \text{NDVI} - 10846.556425 \text{Lat} \cdot \text{Lon} + 5491.635095 \text{Lat}^2 \\
 & - 147.323912 \text{NDVI}^3 + 104.285871 \text{Lon} \cdot \text{NDVI}^2 - 53.848046 \text{Lon}^2 \cdot \text{NDVI} \\
 & - 11.517867 \text{Lon}^3 - 179.190743 \text{Lat} \cdot \text{NDVI}^2 + 123.398379 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\
 & + 52.368359 \text{Lat} \cdot \text{Lon}^2 - 121.157637 \text{Lat}^2 \cdot \text{NDVI} - 25.382096 \text{Lat}^2 \cdot \text{Lon} \\
 & - 25.891741 \text{Lat}^3
 \end{aligned}$$

3.1.5 Wuhan

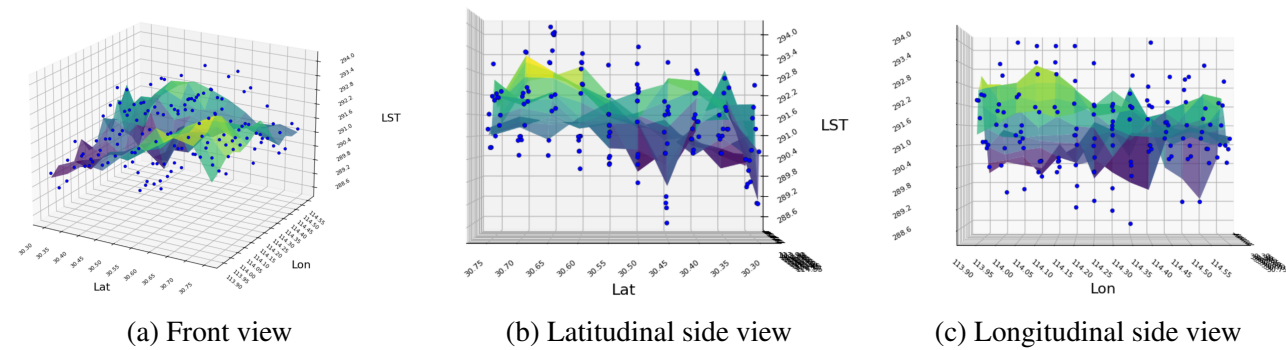


Figure 7: Three views of NDVI-LST Model of Wuhan

For Wuhan, the best-fit model was a 4th-degree polynomial, which achieved an adjusted R_{adj}^2

of 0.3063. This relatively low value suggests that only about 30.6% of the variation in land surface temperature (LST) was explained by the model's predictors, indicating a weaker but still present relationship between LST and variables such as NDVI, longitude, and latitude in this region. The increased degree of the polynomial reflects the need for greater model flexibility to capture the complex interactions specific to Wuhan's urban and environmental characteristics. However, the modest R_{adj}^2 also implies the potential influence of other unmodeled factors or spatial heterogeneity within the city.

$$\begin{aligned}
LST = & -390.579364 - 152.659251 \text{ NDVI} + 6.253376 \text{ Lon} - 41.093846 \text{ Lat} \\
& - 29.009818 \text{ NDVI}^2 - 5812.287704 \text{ Lon} \cdot \text{NDVI} + 476.886255 \text{ Lon}^2 \\
& - 1553.308040 \text{ Lat} \cdot \text{NDVI} - 1498.732521 \text{ Lat} \cdot \text{Lon} - 845.239586 \text{ Lat}^2 \\
& - 2.883710 \text{ NDVI}^3 + 102.349265 \text{ Lon} \cdot \text{NDVI}^2 + 99.563211 \text{ Lon}^2 \cdot \text{NDVI} \\
& - 2.206235 \text{ Lon}^3 - 75.076262 \text{ Lat} \cdot \text{NDVI}^2 + 1.079502 \text{ Lat} \cdot \text{Lon} \cdot \text{NDVI} \\
& - 41.238651 \text{ Lat} \cdot \text{Lon}^2 + 122.482940 \text{ Lat}^2 \cdot \text{NDVI} + 261.383212 \text{ Lat}^2 \cdot \text{Lon} \\
& - 299.545265 \text{ Lat}^3 + 4823.467300 \text{ NDVI}^4 + 338.018399 \text{ Lon} \cdot \text{NDVI}^3 \\
& + 13.700590 \text{ Lon}^2 \cdot \text{NDVI}^2 - 11.722291 \text{ Lon}^3 \cdot \text{NDVI} - 3.166150 \text{ Lon}^4 \\
& - 1426.009997 \text{ Lat} \cdot \text{NDVI}^3 - 120.854765 \text{ Lat} \cdot \text{Lon} \cdot \text{NDVI}^2 \\
& + 126.770304 \text{ Lat} \cdot \text{Lon}^2 \cdot \text{NDVI} + 47.709388 \text{ Lat} \cdot \text{Lon}^3 \\
& + 252.241331 \text{ Lat}^2 \cdot \text{NDVI}^2 - 474.309510 \text{ Lat}^2 \cdot \text{Lon} \cdot \text{NDVI} \\
& - 267.907247 \text{ Lat}^2 \cdot \text{Lon}^2 + 589.165176 \text{ Lat}^3 \cdot \text{NDVI} + 666.811939 \text{ Lat}^3 \cdot \text{Lon} \\
& - 622.650540 \text{ Lat}^4
\end{aligned}$$

$$\begin{aligned}
R &= 3 \\
R_{adj}^2 &= 0.2857
\end{aligned}$$

$$\begin{aligned}
LST = & -49412.450451 - 1870321.546042 \text{ NDVI} + 84104.511126 \text{ Lon} - 292709.274605 \text{ Lat} \\
& + 42226.076466 \text{ NDVI}^2 + 31148.168315 \text{ Lon} \cdot \text{NDVI} - 1333.215941 \text{ Lon}^2 \\
& + 5144.150294 \text{ Lat} \cdot \text{NDVI} + 4159.288346 \text{ Lat} \cdot \text{Lon} + 1826.441778 \text{ Lat}^2 \\
& + 66.961608 \text{ NDVI}^3 - 444.803454 \text{ Lon} \cdot \text{NDVI}^2 - 128.451836 \text{ Lon}^2 \cdot \text{NDVI} \\
& + 5.068105 \text{ Lon}^3 + 279.310623 \text{ Lat} \cdot \text{NDVI}^2 - 49.670939 \text{ Lat} \cdot \text{Lon} \cdot \text{NDVI} \\
& - 11.889602 \text{ Lat} \cdot \text{Lon}^2 + 4.897422 \text{ Lat}^2 \cdot \text{NDVI} - 23.615603 \text{ Lat}^2 \cdot \text{Lon} \\
& + 9.289480 \text{ Lat}^3
\end{aligned}$$

3.1.6 Xiaogan

For Xiaogan, the optimal polynomial degree for the regression model was 3, yielding an adjusted coefficient of determination (R_{adj}^2) of 0.5205. This indicates that about 52.1% of the variance in land

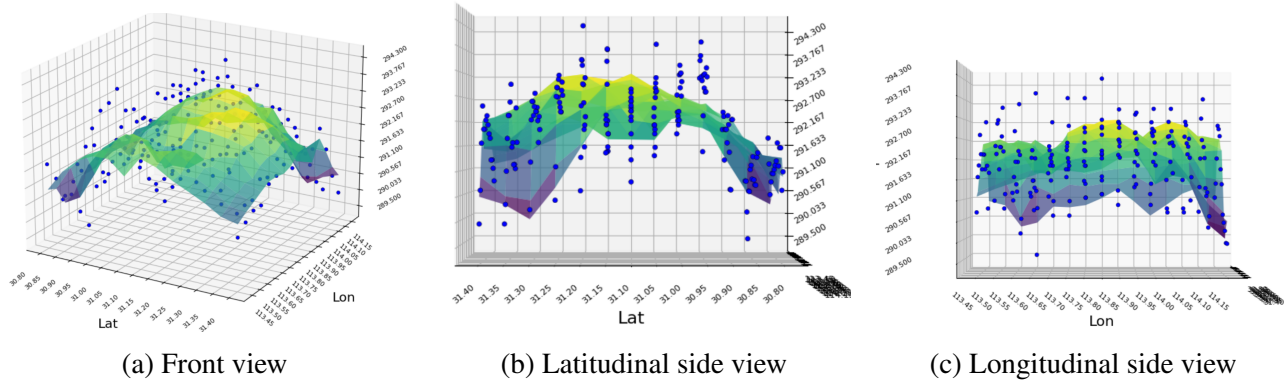


Figure 8: Three views of NDVI-LST Model of Xiaogan

surface temperature (LST) was explained by the predictors, suggesting a moderate fit.

$$\begin{aligned}
 \text{LST} = & 4744.626068 - 541669.894922 \text{ NDVI} + 42498.614498 \text{ Lon} - 151129.634183 \text{ Lat} \\
 & - 8277.229968 \text{ NDVI}^2 + 7730.368371 \text{ Lon} \cdot \text{NDVI} - 853.396099 \text{ Lon}^2 \\
 & + 6788.791504 \text{ Lat} \cdot \text{NDVI} + 3439.489336 \text{ Lat} \cdot \text{Lon} - 1456.279693 \text{ Lat}^2 \\
 & + 78.653745 \text{ NDVI}^3 + 62.915636 \text{ Lon} \cdot \text{NDVI}^2 - 22.676593 \text{ Lon}^2 \cdot \text{NDVI} \\
 & + 3.780204 \text{ Lon}^3 + 33.088169 \text{ Lat} \cdot \text{NDVI}^2 - 84.355991 \text{ Lat} \cdot \text{Lon} \cdot \text{NDVI} \\
 & - 13.865606 \text{ Lat} \cdot \text{Lon}^2 + 44.532645 \text{ Lat}^2 \cdot \text{NDVI} - 4.076154 \text{ Lat}^2 \cdot \text{Lon} \\
 & + 20.238648 \text{ Lat}^3
 \end{aligned}$$

3.1.7 Wuxi

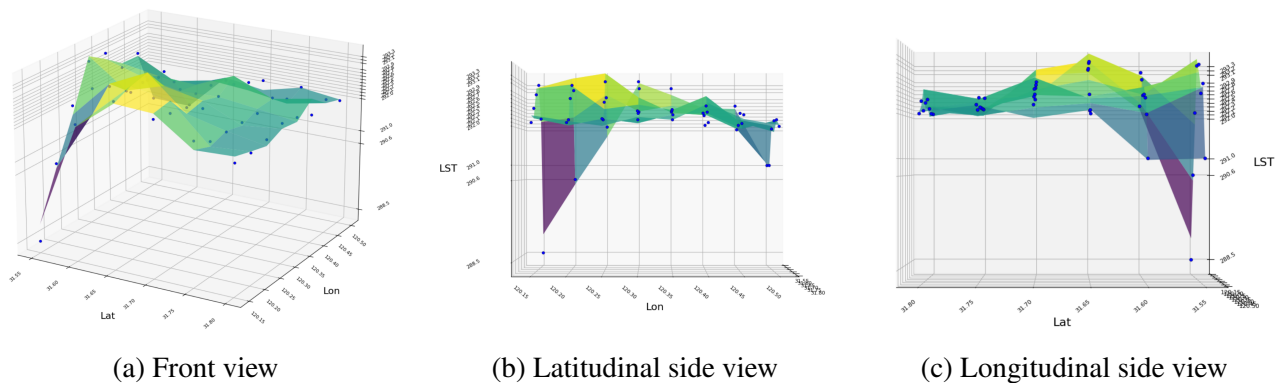


Figure 9: Three views of NDVI-LST Model of Wuxi

For Wuxi, the best-fitting model was a 3rd-degree polynomial, achieving an adjusted coefficient

of determination of 0.7859, indicating a relatively strong fit where nearly 78.6% of the variance in LST is explained by the model. However, due to the land area of Wuxi being divided into two separate parts, the available sample size was quite limited. This limitation led to overfitting issues when fitting models with degrees higher than 5, where the R_{adj}^2 values exceeded 1, which is statistically invalid and a clear sign of overfitting. Therefore, restricting the model degree to 3 provided a balance between capturing complex relationships and maintaining model validity. The equation provided as follow:

$$\begin{aligned} \text{LST} = & -22270.446943 + 1092850.927488 \text{NDVI} - 1800364.152000 \text{Lon} + 6830924.201543 \text{Lat} \\ & + 233193.441218 \text{NDVI}^2 - 41749.407451 \text{Lon} \cdot \text{NDVI} + 20523.490344 \text{Lon}^2 \\ & + 84344.571400 \text{Lat} \cdot \text{NDVI} - 41855.680053 \text{Lat} \cdot \text{Lon} - 136549.567422 \text{Lat}^2 \\ & + 1047.584166 \text{NDVI}^3 - 1992.917338 \text{Lon} \cdot \text{NDVI}^2 + 190.754369 \text{Lon}^2 \cdot \text{NDVI} \\ & - 55.912860 \text{Lon}^3 + 176.790215 \text{Lat} \cdot \text{NDVI}^2 - 86.573819 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\ & - 12.460895 \text{Lat} \cdot \text{Lon}^2 - 1168.591376 \text{Lat}^2 \cdot \text{NDVI} + 708.094449 \text{Lat}^2 \cdot \text{Lon} \\ & + 544.428491 \text{Lat}^3 \end{aligned}$$

3.1.8 Huzhou

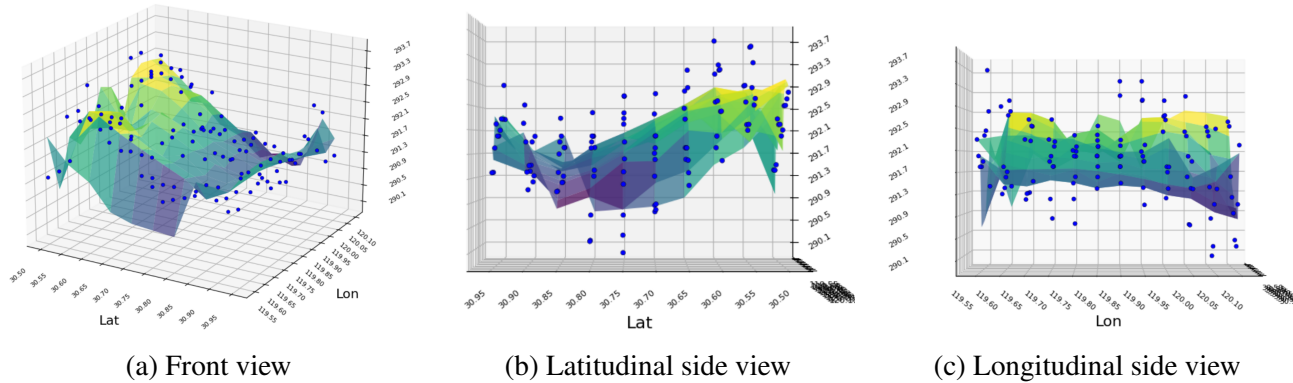


Figure 10: Three views of NDVI-LST Model of Huzhou

For Huzhou, a 3rd-degree polynomial regression model was identified as the best fit, explaining approximately 48.9% of the variability in land surface temperature (LST), as indicated by an adjusted

R^2 of 0.4889.

$$\begin{aligned} \text{LST} = & -5344.141271 - 413585.594116 \text{NDVI} - 62209.090221 \text{Lon} + 248404.900020 \text{Lat} \\ & - 2903.515120 \text{NDVI}^2 + 3570.878194 \text{Lon} \cdot \text{NDVI} + 451.894749 \text{Lon}^2 \\ & + 13079.355238 \text{Lat} \cdot \text{NDVI} + 494.835721 \text{Lat} \cdot \text{Lon} - 9162.306923 \text{Lat}^2 \\ & - 90.020396 \text{NDVI}^3 + 20.034278 \text{Lon} \cdot \text{NDVI}^2 - 9.413802 \text{Lon}^2 \cdot \text{NDVI} \\ & - 0.819667 \text{Lon}^3 + 20.348286 \text{Lat} \cdot \text{NDVI}^2 - 43.341540 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\ & - 5.070566 \text{Lat} \cdot \text{Lon}^2 - 128.680957 \text{Lat}^2 \cdot \text{NDVI} + 12.025854 \text{Lat}^2 \cdot \text{Lon} \\ & + 84.594600 \text{Lat}^3 \end{aligned}$$

3.2 Quantitative Comparison among Cities

Across the eight cities analyzed, both the adjusted R_{adj}^2 values and best-fitting polynomial degrees varied, reflecting underlying differences in how effectively NDVI and spatial variables (latitude and longitude) explain variations in land surface temperature (LST). These differences provide insight into each city's land-use composition, vegetation cover, and urban structure.

Jiaying stood out with the highest adjusted R^2 of 0.8468, suggesting a very strong relationship between LST and the predictor variables. This high explanatory power may indicate a relatively uniform urban layout and consistent vegetation patterns, which the model captures well using a degree 3 polynomial. Likewise, Wuxi and Shanghai achieved high adjusted R^2 values of 0.6935 and 0.6219, respectively. These results point to effective model performance, though in the case of Wuxi, the use of a degree 4 polynomial and the observation that higher-degree models produced implausibly high R_{adj}^2 values (greater than 1) suggest overfitting, likely due to the limited sample coverage caused by fragmented land boundaries.

On the other end of the spectrum, Wuhan (0.3063) and Suzhou (0.3688) exhibited relatively low adjusted R^2 values, indicating weaker model performance. This may reflect greater heterogeneity in land cover types, uneven distribution of green space, or other unmodeled influences like elevation or impervious surfaces. For Wuhan in particular, the requirement for a degree 4 polynomial despite the low R_{adj}^2 suggests that the LST patterns are complex but not well-explained by NDVI and spatial coordinates alone.

In terms of polynomial complexity, most cities reached optimal model performance with a degree 3 polynomial, indicating that moderate nonlinear interactions between NDVI, latitude, and longitude sufficiently captured the trends. However, Wuhan and Wuxi required degree 4, reflecting either more complex temperature-vegetation relationships or possibly a need to account for subtle interactions not captured by simpler models.

Overall, these results suggest that urban form and vegetation-temperature relationships vary significantly across cities. Cities with compact or well-vegetated cores (like Jiaxing) may exhibit more predictable cooling patterns due to vegetation, while more sprawling or heterogeneous cities (like Wuhan) show more diffuse or inconsistent trends. These insights highlight the value of using city-specific modeling approaches when analyzing environmental patterns, rather than assuming a one-size-fits-all model.

3.3 Geographical Patterns Analysis

3.4 Coefficient Patterns Analysis